

Simulation of Partially Molten Materials

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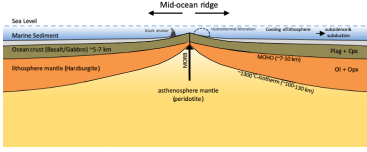
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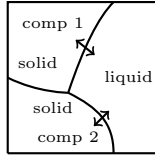
Partial melting in mantle



1. Generates the oceanic and continental crust.
2. Determines the large-scale chemical evolution of the Earth.

Governing equations for physics

Two phase system : Immiscible solid mixture melting into miscible liquid.



Fluid Mechanics : High-viscosity Stokes flow governs

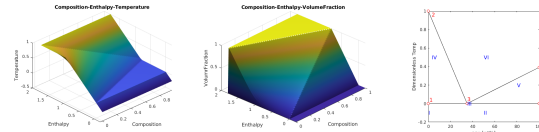
the mantle (3), while magma follows Darcy's law (1), alongside continuity (2) and compaction equations (4).

$$\begin{aligned}
 (1) \quad & \phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k\phi_l}{\mu l} \nabla(p_l - \rho_f \mathbf{g}) = 0, \\
 (2) \quad & \nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \\
 (3) \quad & \nabla \bar{p} - \nabla \cdot \tau(\mathbf{v}_s) = (\rho_s - \rho_l) \mathbf{g}, \\
 (4) \quad & \mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0,
 \end{aligned}$$

Transport : Trace the bulk composition of one specie (5) and transport bulk energy with enthalpy method (6).

$$\begin{aligned}
 (5) \quad & \frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi_l \mathcal{D}_l \nabla c_l) = 0. \\
 (6) \quad & \frac{\partial H}{\partial t} + \nabla \cdot \bar{\mathbf{v}} T = \frac{L}{c_p} \nabla \cdot \phi_s \mathbf{v}_s + \kappa \nabla^2 T + \frac{\alpha_p}{c_p} T \bar{\mathbf{v}} \cdot \mathbf{g},
 \end{aligned}$$

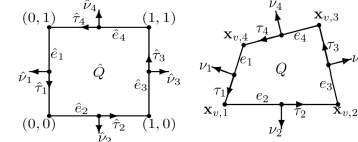
Phase change : Assume thermodynamic equilibrium achieves at each time step. Approximated with eutectic phase behavior.



Numerical methods

Various numerical methods are adapted and implemented to handle the strong degeneracy and discontinuities.

Spatial discretization : Quadrilaterals



Mixed finite element method : Scaled weak form accounting for singularity induced by zero porosity. (Arbogast, Hesse and Taicher, 2017)

BDM element: $\mathbb{V}_D \times \mathbb{W}_D \subset H_0(\text{div}; \Omega) \times L^2(\Omega)$. Lowest order reduced $H(\text{div})$ -conforming mixed space. (Arbogast, Tao and Wang[4]).

$$\begin{aligned}
 \mathbb{V}_{D,h}(Q) &= \mathbb{V}_1^0(Q) = \mathbb{P}_1^2 \oplus \text{curl } \mathbb{S}_2^{\mathcal{DS}}(Q) \\
 \mathbb{W}_{D,h}(Q) &= \mathbb{W}_1^0(Q) = \mathbb{P}_0,
 \end{aligned}$$

Accuracy $\mathcal{O}(h^2)$ and $\mathcal{O}(h)$. **DOF** 8 + 1 = 9.

Modified BR element: $\mathbb{V}_S \times \mathbb{W}_S \subset (H_0^1(\Omega))^2 \times L^2(\Omega)$

$$\begin{aligned}
 \mathbb{V}_{S,h}(Q) &= \mathcal{DS}_1^2(Q) \oplus \text{span}\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4\} \\
 \mathbb{W}_{D,h}(Q) &= \mathbb{W}_1^0(Q) = \mathbb{P}_0,
 \end{aligned}$$

Accuracy $\mathcal{O}(h^2)$ and $\mathcal{O}(h)$. **DOF** 12 + 1 = 13.

Saddle point system: Resulted saddle point system will be solved with modified Uzawa iteration.

Multi-level WENO method : New nonlinear weighting scheme, (Arbogast, Huang and Tian, 2024)

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{s r_l + \eta_l}}$$

The biasing term η_l is defined as following to bias away from small stencils when solution is smooth:

$$\eta_l = \begin{cases} 1, & \text{if } r_l = 1, \\ 3, & \text{if } r_l = 2, \\ 4, & \text{if } r_l \geq 3, \end{cases}$$

The normalized nonlinear weights are then:

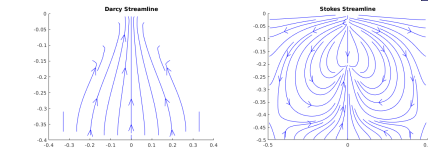
$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_k \hat{\omega}_k}$$

Code implementation

1. Iterative linear solver provided by PETSc.
2. Direct solver provided by LAPACK.
3. Parallelization provided by PETSc and MPI.

Numerical results

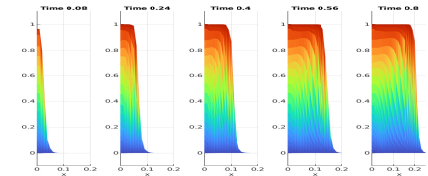
Mid ocean ridge corner flow



MLWENO boundary treatment

Inflow Stabilization : Burgers-like equation $u_t + u^3 u_x = 0$ with the flux $\mathbf{f}(u) = (u^4/4, 0)$ in the domain $(0, 1)^2$.

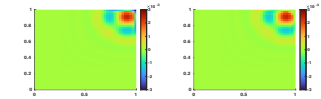
$$u_D(0, y, t) = \begin{cases} \sin^2(2\pi y), & y \leq 0.25, y \geq 0.75 \\ 1, & 0.25 < y < 0.75 \end{cases}$$



ML-WENO(3,2,1) at inflow.

Outflow error elimination : sine wave traveling diagonally across the domain $(0, 1)^2$ with the exact solution:

$$u(x, y, t) = \sin^6\left(\frac{x - 0.1 - t}{0.8}\pi\right) \sin^6\left(\frac{y - 0.1 - t}{0.8}\pi\right), 0.1 + t \leq x, y \leq 0.9 + t$$



Column melt extraction : SSPRK2

